



RELATIONS – EXAMPLES

CMSC 56 – DISCRETE MATHEMATICS IN
COMPUTER SCIENCE I

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Major Reference Books:

Kenneth H. Rosen (2012). Discrete Mathematics and Its Applications. 7th Edition. The McGraw-Hill Companies, Inc.

K.H. Rosen, J.G. Michaels, J.L. Gross, J.W. Grossman, D.R. Shier (2000). Handbook of Discrete and Combinatorial Mathematics, CRC Press LLC



At the end of this lecture, the students should be able to:

- Recall properties of binary relations
- Define equivalence relations, partial orders and total orders
- Apply the definitions to determine the type of a given relation.

Recall

Here are four basic properties of a binary relation:

- Reflexive - if xRx for all x in A
- Symmetric - if for all $x, y \in A$, xRy implies yRx
- Anti-symmetric - if for all $x, y \in A$, xRy and yRx imply that $x = y$
- Transitive - if for all $x, y, z \in A$, xRy and yRz imply that xRz

Reflexive Relations

Example:

- The relation $\leq$ on the set of integers $\{1,2,3\}$ is reflexive
- The relation $\subseteq$ on any subset of a universal set is reflexive
- The relation $>$ or $<$ is not reflexive

A set is called **irreflexive** if and only if for all $x \in A$, x is not related to itself.

Symmetric Relations

Example:

- The relation $=$ on the set of integers $\{1,2,3\}$ is symmetric
- The relation "*being acquainted to*" is symmetric.
- The relation $\leq$ or $\geq$ is not symmetric.

Antisymmetric Relations

Example:

- The relation $>$ or $<$ is antisymmetric
- The relation $\leq$ or $\geq$ is antisymmetric.

Transitive Relations

Example:

- The relation $\leq$ or $\geq$ is transitive.
- The relation $=$ is transitive

Relations and Directed Graphs

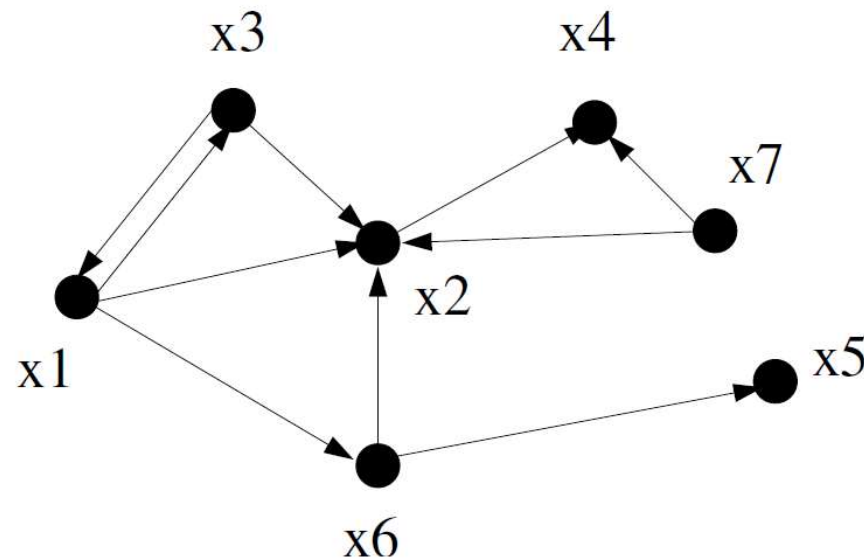
A relation can be modeled with a directed graph $G = (V, E)$ with vertex set $V = A$ and edge set $E = R$.

- **Directed graphs**, called **digraphs** for short, represent how things are connected together and how to get from one thing to another by following those connections.
 - Usually pictured as bunch of dots/circles (**nodes or vertices**) with the lines connecting them (**called directed edges or arrows**).

Digraphs

Example:

Let the vertices $x_1, x_2, \dots, x_n$ correspond to web pages. The arrows are used to indicate when one page has a hyperlink to another:



In 1995, two students at Stanford, Larry Page and Sergey Brin, ultimately became multibillionaires from the realization of how useful the structure of this graph in building a search engine.

Digraphs

Determine the properties of the following relations illustrated by digraphs:



(a)



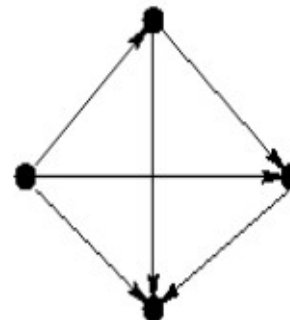
(b)



(c)



(d)



(e)

Digraphs

Determine the properties of the following relations illustrated by digraphs:

Answer:

a) Reflexive,
symmetric,
anti-symmetric

b) Symmetric,
anti-symmetric

c) Irreflexive

d) Irreflexive,
symmetric

e) Irreflexive,
asymmetric,
transitive



(a)



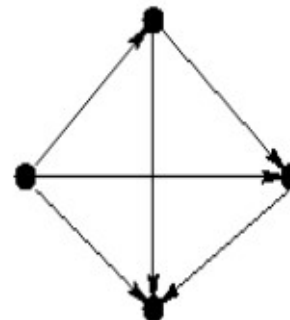
(b)



(c)



(d)

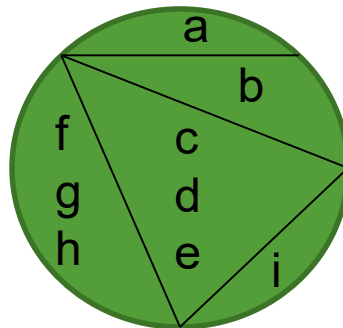


(e)

Partitions

Let S be a nonempty set. A partition of S is a subdivision of S into non-overlapping, nonempty subsets. Precisely, a partition of S is a collection $P = \{A_i\}$ of nonempty subsets of S such that

- i. Each $a \in S$ belongs to one of A_i .
- ii. The sets $\{A_i\}$ are mutually disjoint.



The green circle = $\{a, b, c, d, e, f, g, h, i\}$ is partitioned into 5 parts.

Partitions

The subsets in a partition are called **cells**.
Thus each $a \in S$ belongs to exactly one of the cells.

Equivalence Relation

Definition:

A binary relation R on a set S is an **equivalence relation** if and only if R is reflexive, symmetric and transitive.

Example:

- The equality relation is an equivalence relation.

Equivalence Relations

Example: Consider the relation R on $S = \{1,2,3,4\}$ given by

$$R = \{(1,1), (2,2), (1,3), (3,1), (3,3), (4,4)\}.$$

Is R an equivalence relation?

Equivalence Relations

Example: Consider the relation R on $S = \{1,2,3,4\}$ given by

$$R = \{(1,1), (2,2), (1,3), (3,1), (3,3), (4,4)\}.$$

Is R an equivalence relation?

Yes, R is reflexive, symmetric and transitive.

Order Relations

Several types of relations allow us to rank different objects against one another.

These relations will allow us to say that some objects are “greater” or “less” than others, to decide what objects are “better” or “worse” than another, etc.

Strict Order

Consider a non-empty set S . A relation R on S is a **strict order** of S if it is irreflexive, asymmetric and transitive (such as the letter e digraph example).

Common examples: $<$, $>$, $\subset$

Partial Ordering Relations

Consider a non-empty set S . A relation R on S is a **partial ordering** of S or a partial order on S if it is reflexive, antisymmetric and transitive. (Can you revise the letter e digraph example to represent this?)

The ordered pair (A, R) is called a **poset** (partially ordered set) when R is a partial order on A .

Partial Ordering Relations

Examples:

- The relation $\subseteq$ defined over set A
- The relation $\leq, \geq$ defined over the set of real numbers

Partial Ordering Relations

Example:

The relation “is an ancestor of” on the set of all human beings that have ever lived.

- (Reflexive) I am an ancestor of myself.

Partial Ordering Relations

Example:

The relation “is an ancestor of” on the set of all human beings that have ever lived.

- (Antisymmetric) My father is an ancestor of me, but I’m not an ancestor of my father.

Partial Ordering Relations

Example:

The relation “is an ancestor of” on the set of all human beings that have ever lived.

- (Transitive) My grandfather is an ancestor of my father, and my father is an ancestor of me, so my grandfather is an ancestor of me.

Partial Ordering Relations

Example:

The relation “is an ancestor of” on the set of all human beings that have ever lived.

- There might be people who are not ancestors of each other at all, such as my father and mother.

Total Order

A binary relation on a set A is a **total order** if and only if it is

1. a partial order, and
2. for any pair of elements a, b of A , a is related to b or b is related to a .

Every element is related with every element one way or the other. A total order is also called a **linear order**.

Total Order

Example:

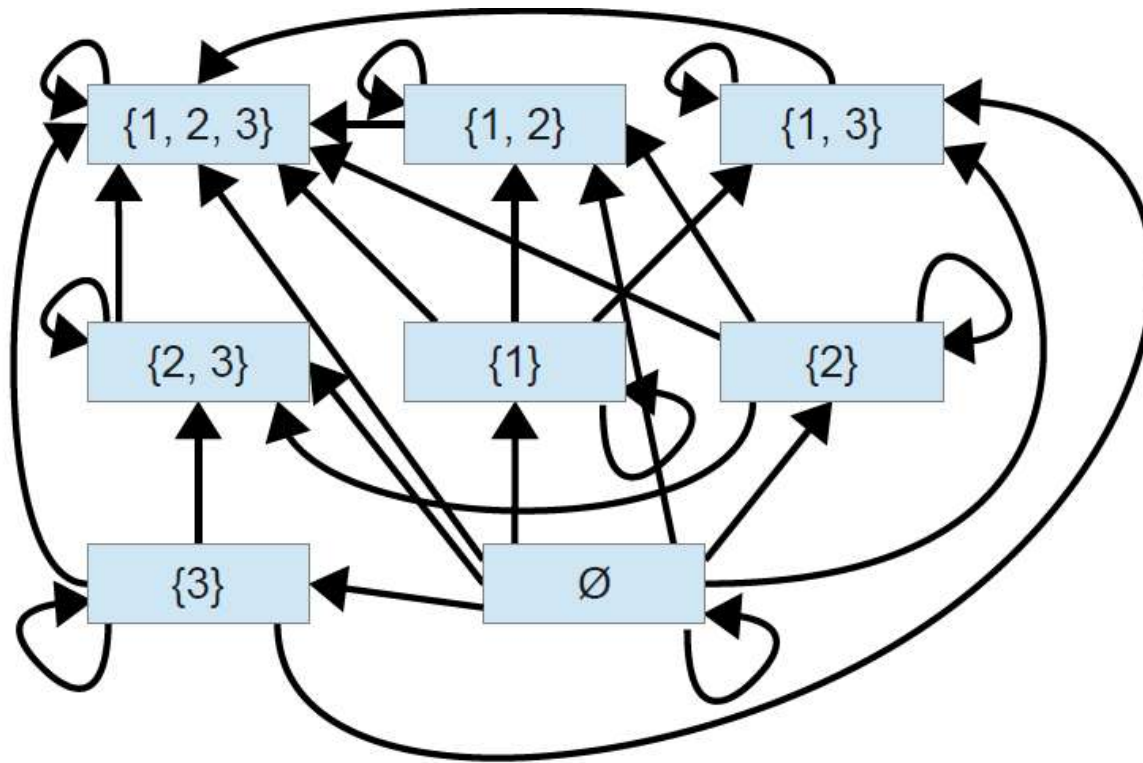
- The relation $\leq$ on the set of integers is a total order.

Illustrations: Hasse Diagrams

Define the relation $\subseteq$ over the power set of $A = \{1,2,3\}$.

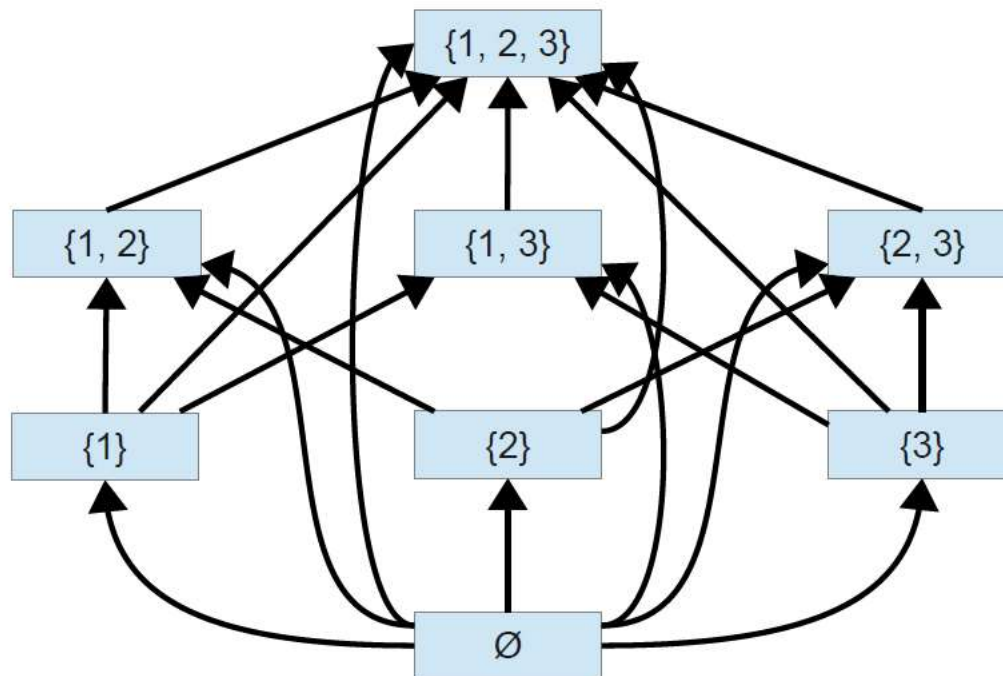
Try to draw a digraph of the relation stated.

Illustrations: Hasse Diagrams



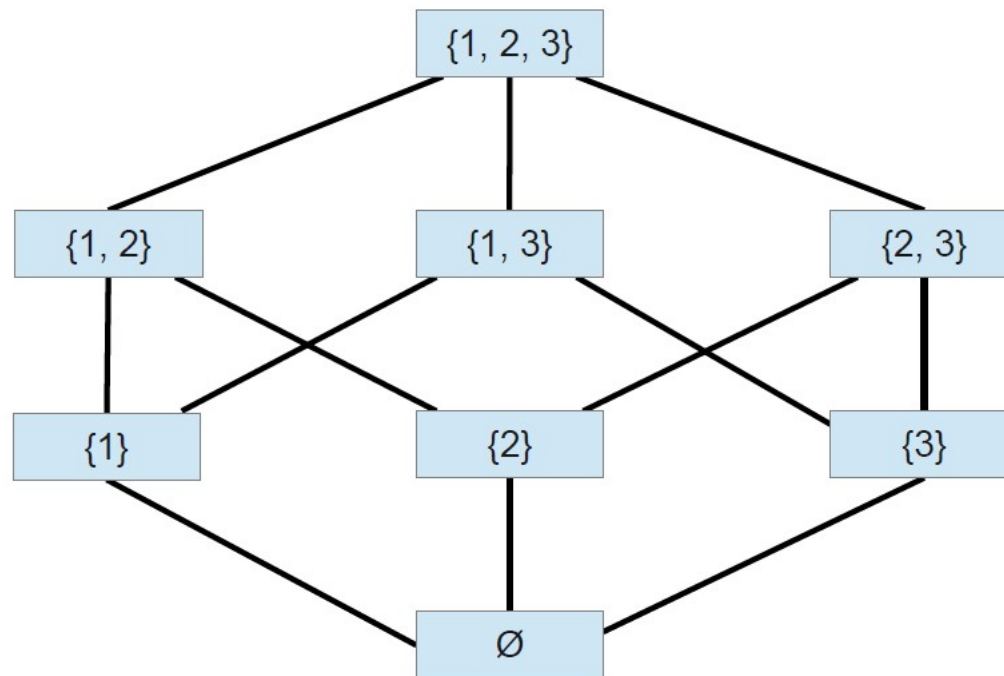
This digraph illustrates the relation $\subseteq$ (subset) over the power set of $A = \{1, 2, 3\}$.

Illustrations: Hasse Diagrams



This digraph is achieved by arranging the sets having the most elements to the least and removing the 'reflexive' part.

Illustrations: Hasse Diagrams



This is the Hasse diagram of the above ordered set.

Illustrations: Hasse Diagrams

Given an order relation R over a set A , a Hasse diagram for R is a drawing of the elements of A with the following properties:

1. All edges are undirected
2. If there is an edge from x to y and x is below y in the diagram, then xRy
3. There are no self-loops
4. There are no redundant edges: if there is an edge from x to y and an edge from y to z , then there is no edge from x to z .

Illustrations: Hasse Diagrams

Here is another example of a Hasse diagram for $<$ relation over the set $\{1,2,3,4\}$

